

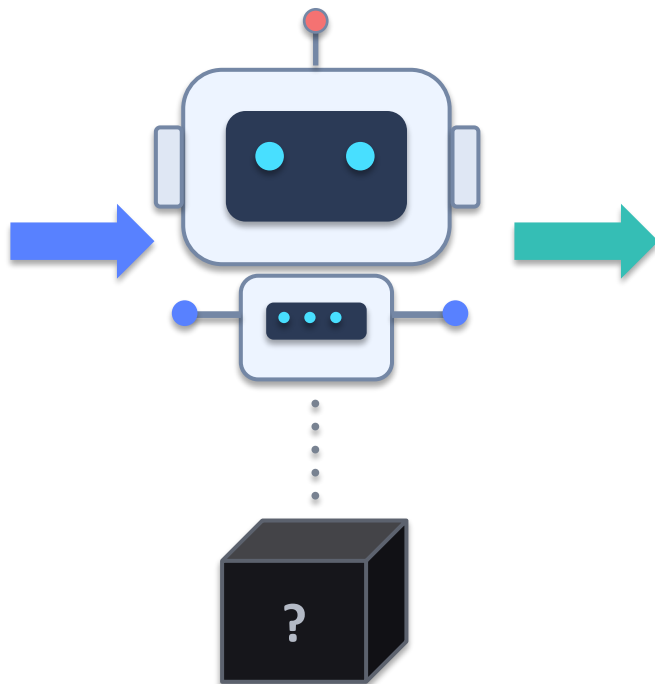
Cognitive-Demand Rubrics as Predictors of Vision-Language Reasoning Failure

Can we predict WHEN a model fails?

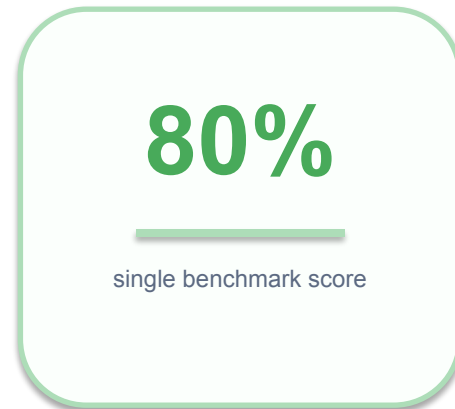
Benchmark



Vision LLM



Accuracy

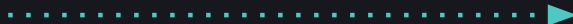
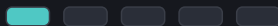


Demand is intrinsic to claim evidence pairs

EASY CLAIM

How many lines?

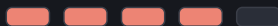
LOW DEMAND



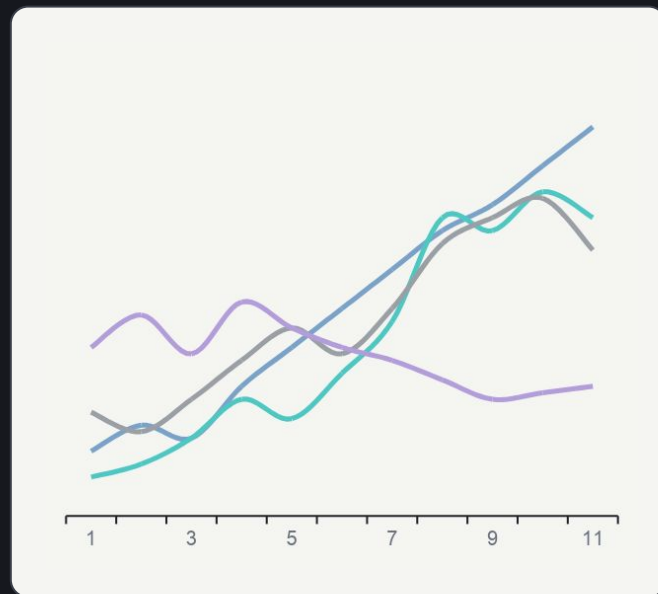
HARD CLAIM

Which line has the greatest AUC?

HIGH DEMAND



EVIDENCE



Three claim-evidence science-reasoning benchmarks

Feature Engineering

SciVer

- 817 validated reasoning tasks

SciClaimEval

- 265 validated reasoning tasks

Test

CharXiv

- 1000 validated reasoning tasks
- Answers for 39 models
(incl. Human, GPT-4o-Blind baselines)

Article

General scales unlock AI evaluation with explanatory and predictive power

<https://doi.org/10.1038/s41586-026-10303-2>

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- **Zhou et al., Nature 2026** — measures what a task demands
- We repurpose the text rubrics for *visual* evidence
- **Our approach:** leverage frontier VLMs (Fable/Opus) to engineer the missing visual demand dimensions, with Zhou et al. as the guide

MCu — Calibrating Knowns and Unknowns

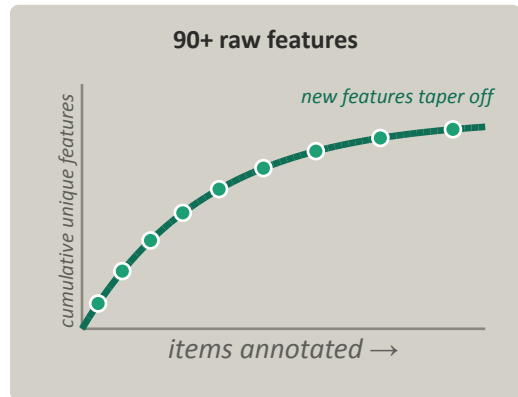
rubric R2, each demand rated 0–5

0 None	Immediately clear whether you know it or not; no self-assessment needed.
1 Very low	Minimal effort; the boundary between knowing and not knowing is very clear.
2 Low	Some effort to check what you know, but the assessment is straightforward.
3 Medium	Moderate effort; you may only think you know rather than truly know.
4 High	Significant effort; the line between knowing and not knowing blurs.
5+ Very high	Sophisticated self-assessment to avoid over- or under-confidence.

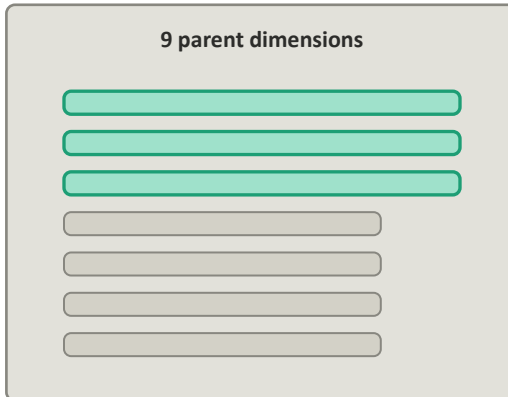
Zhou, L. et al. General scales unlock AI evaluation with explanatory and predictive power. Nature (2026).

Engineering of emergent visual demand dimensions

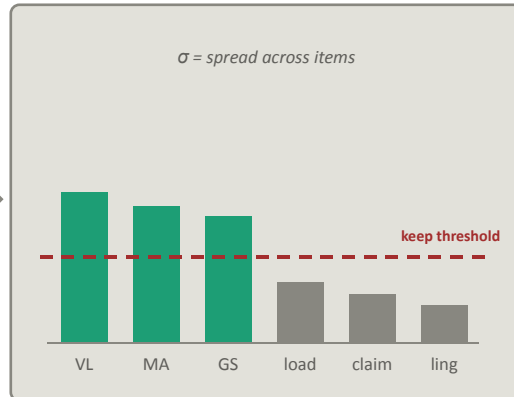
1 · Annotation (Fable/Opus)



2 · Semantic grouping



3 · Spread test (keep what varies)



4 · Survivors

Visual localization

VL — find the right point

Multi-element aggregation

MA — combine multiple data points

Gestalt & shape

GS — read trend / shape

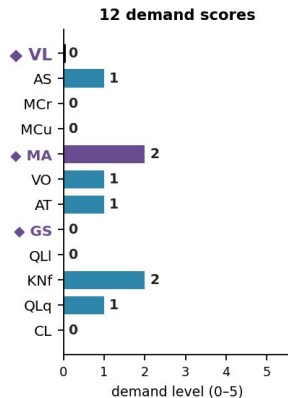
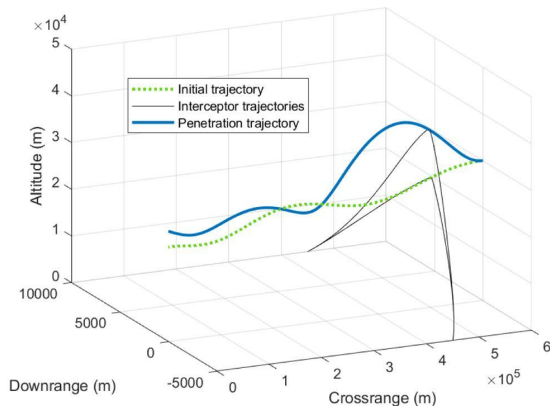
Rubric Summary: Visual Localization (VL)

0 None	Target is the only element and in view.
1 Very low	Single salient target; set apart by a clear label or unique position.
2 Low	Several like candidates; selected by reading a printed distinguisher (panel title, legend, header).
3 Intermediate	Many like candidates read one-by-one, or a region cued by a drawn guide (gridline, quadrant).
4 High	A semantic region the figure never names.
5+ Very high	Exhaustive search, no cue; entity has no label and is found via an inferred external property.

◆ Visual Localization — low vs high demand

Q: How many dimensions are shown in this plot?

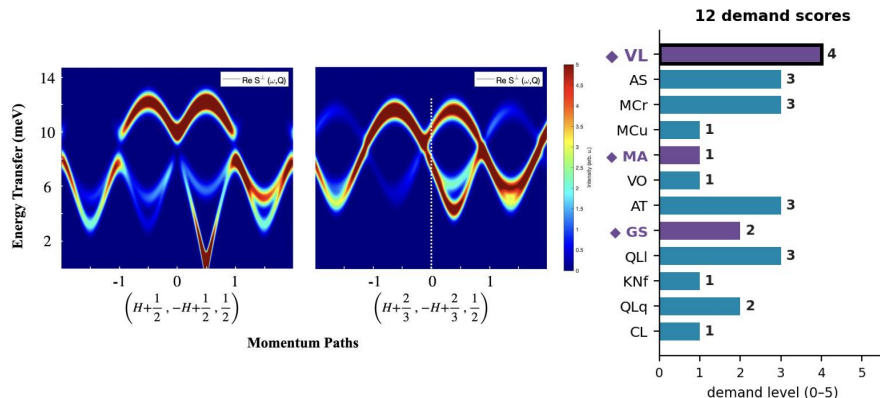
LOW demand · VL = 0



Answer: 3

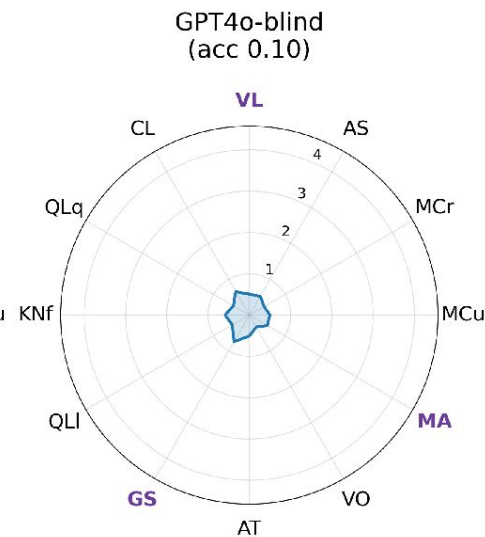
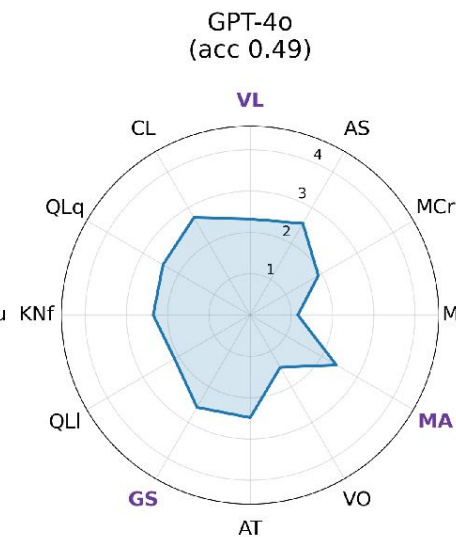
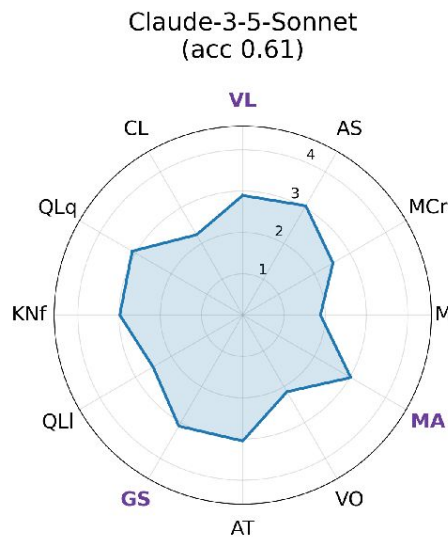
Q: What is the x-axis label of the chart where its corresponding subplot has less area with intensity >4.5 when the meV is between 6 and 10?

HIGH demand · VL = 4



Answer: $\left(H+\frac{1}{2}, -H+\frac{1}{2}, \frac{1}{2}\right)$

Per model ability radars (AUC-ability)



KPIs and feature selection

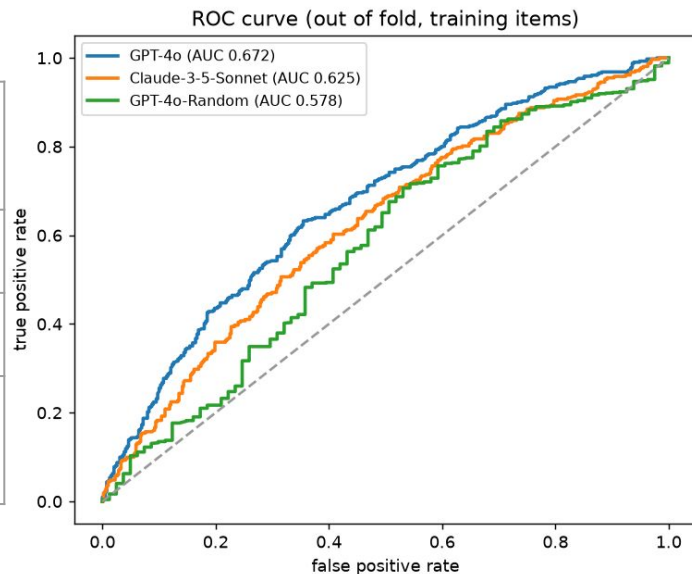
- KPIs: ROC-AUC (primary, threshold invariant); balanced accuracy and F1 (secondary, threshold dependent)
- Metadata (publication year, subject, etc.) serves as weak input baseline for classifiers; dropped as input features for final model
- Twelve demand features substantially increase model performance; one (CL—conceptualization, learning, and abstraction) dropped for having mostly zero values

Input features	ROC-AUC (GPT-4o classifier)	ROC-AUC (Sonnet 3.5 classifier)
Metadata	0.53	0.54
Demand dims	0.68	0.63
Metadata + demand dims	0.67	0.62

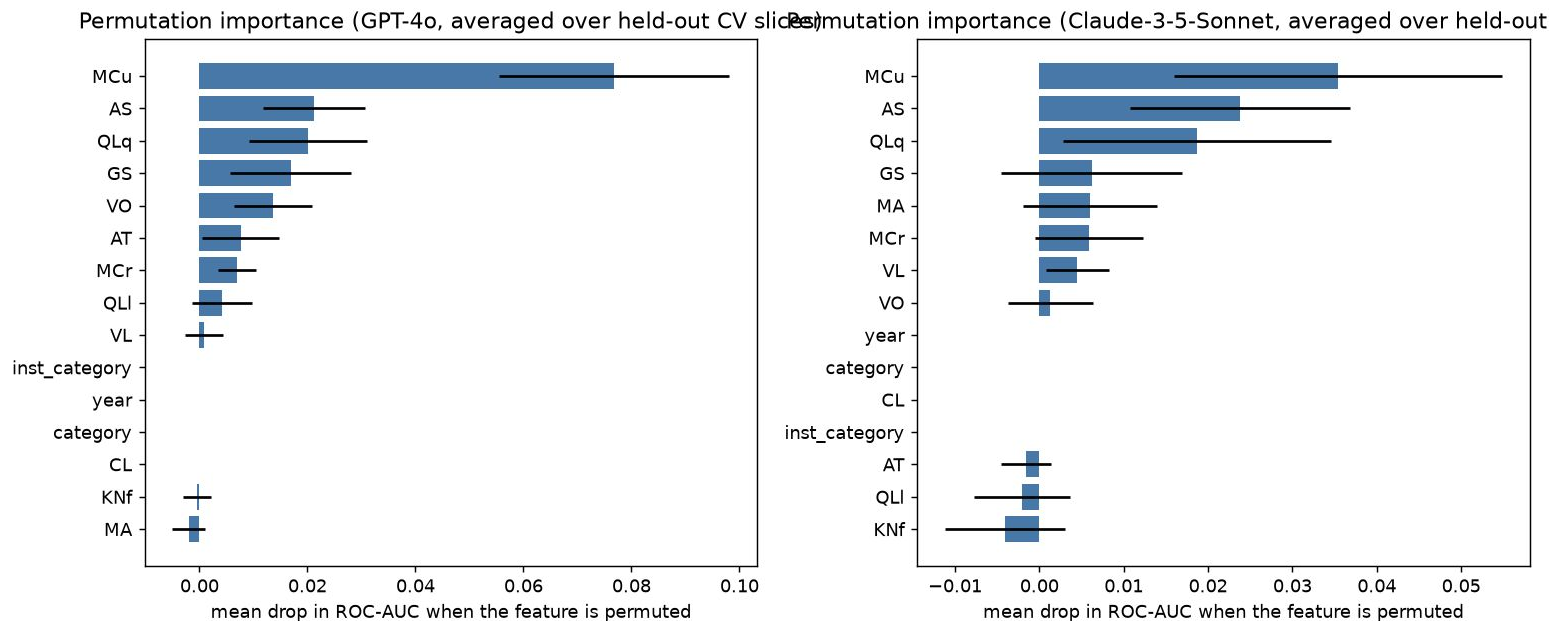
Modeling journey and cross-validation performance

- Fitted and tuned logistic, random forest, histogram gradient boosting, and XGBoost classifiers. Evaluated using CV on original training set and 30 bootstrap resamples.
- Logistic regression was chosen as the final model since more complicated models provided no clear performance boost.

Target	ROC-AUC (CV)	Balanced accuracy	F1
GPT-4o	0.67	0.63	0.65
Sonnet 3.5	0.63	0.57	0.38
GPT-4o random	0.58	0.50	0.95

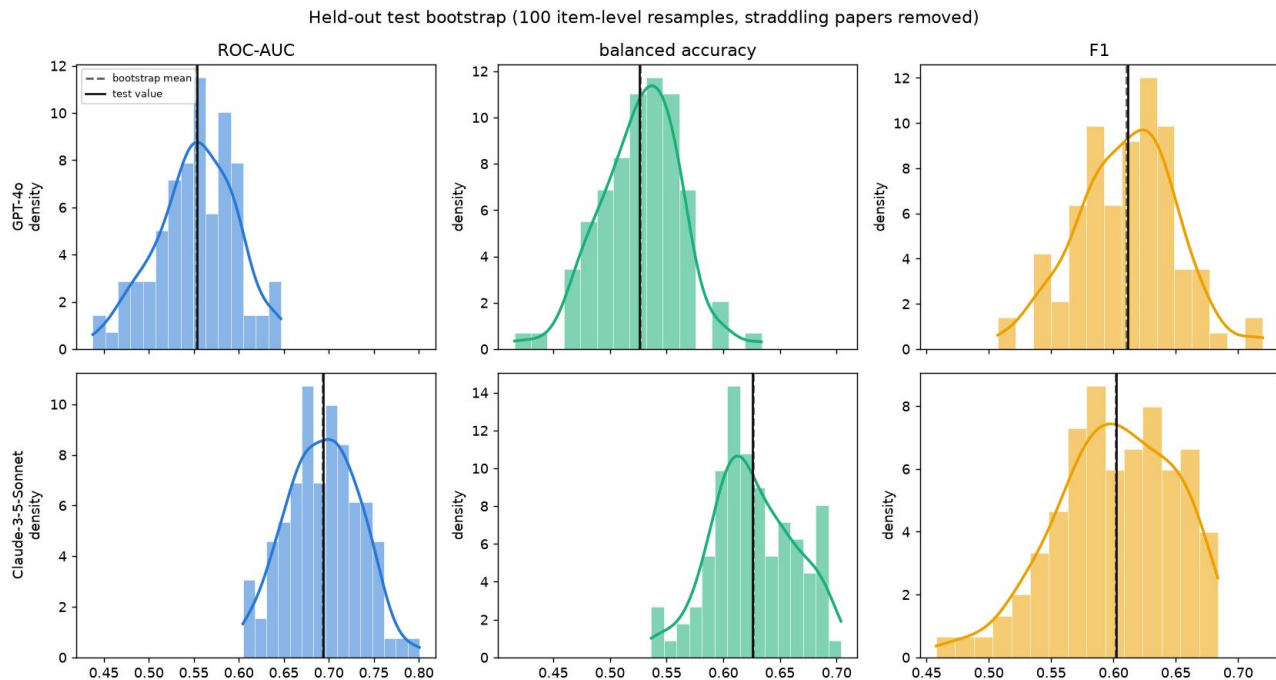


Performance drivers



MCu (calibrating knowns and unknowns), AS (attention and scan), QLq (quantitative reasoning, and GS (gestalt and shape judgement) are the most informative demand dimensions for our models.

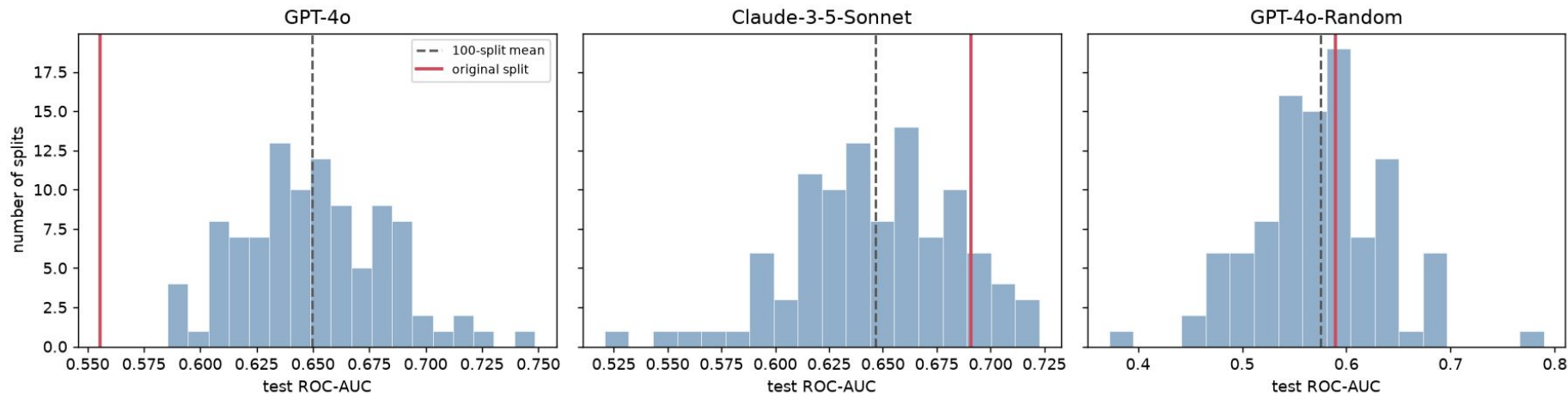
Test set performance



KPIs on test set and 100 bootstrap samples. The GPT-4o classifier performs worse than expected across all bootstrap samples, while the Sonnet 3.5 classifier performs better.

Post hoc audit on train-test split sensitivity

Test ROC-AUC over 100 random paper-grouped splits (original split marked)



KPIs on 100 random train-test splits. In particular, the GPT-4o classifier's average performance is much better across the additional train-test splits.

Limitations and next steps

- Annotated CharXiv dataset is small
- Further investigation should include data with more
 - More claim-evidence pairs
 - Subject representation
 - Rubric types (textual, visual, equation/mathematical)
 - Target VLMs

Conclusions and decision relevance

- Demand scores modestly predict VLM failure.
- Emergent GS dimension carries informative signal.
- Assist with routing to frontier VLMs.
- Model-agnostic rubrics can transfer to new datasets and target models.